

Logging, Detection, Windows Security, and Basic Threat Hunting

These notes were written by Kevin Beideman. The recommended time to understand and practice these concepts are 2 hours and 15 minutes.

Windows Security Logging Deep Dive

Lock/unlock PC, Open/close apps, Intentionally fail a login once, Launch Powershell, Connect to websites, run a ping, run an Nmap scan from Kali

Open Event Viewer. Windows + R: eventvwr.msc

What events are generated during login?

1. *Authentication Event: system checking whether the username and password are correct*
 - a. *Event ID 4624 -> Successful Login*
 - b. *Event ID 4625 -> Failed Login*
2. *Account Validation*
3. *Session Creation*
4. *Logon Type Event: records how the user logged in*
5. *Process Creation Events*
6. *More but we won't address them now*

What logs track process creation?

Operating system event logs. The main log is: Security Event Log. Event ID 4688 is Process Creation.

What is Windows auditing?

Security feature in Windows that records important system and user activities into event logs so administrators, security analysts, and investigators can monitor what happens on a computer or network.

Filter the logs for: Warning, Error, or Information. Search for 4624 and 4625. Document timestamps, usernames, logon types, source processes.

4624: 5/27/2026 12:09:21 PM, Username: REDACTED for security purposes, Logon Type: 5, Source Processes: services.exe in System32 folder. Logon Type 5 is a service started/logged on.

4625: 5/27/2026 12:04:03 PM, Username:, Logon Type 2, Source Process: svchost.exe in System32. Logon type 2 is an interactive logon (using keyboard, mouse, physical console).

What is Event ID 4688?

A new process has been created.

Identify: PowerShell launches. Cmd launches. Browser launches.

Powershell + Suspicious Activity Detection

What is PowerShell logging?

A security and auditing feature in PowerShell that records PowerShell activity so administrators and security analysts can monitor what commands and scripts were executed on a system.

What is Script Block Logging?

An advanced PowerShell security feature that records the actual PowerShell code being executed on a system.

Why do attackers abuse PowerShell?

It is extremely powerful, built into Windows, trusted by the OS, and capable of controlling almost every part of a computer. It allows attackers to perform malicious actions while blending in with normal administrator activity.

Run commands: **Get-Process, Get-Service, Get-NetTCPConnection, ipconfig, whoami**

Search logs for: powershell

What are command-line arguments?

*Extra pieces of information passed to a program when it is launched from a command line or script. They tell the program: what to do, what options to use, and what file or target to operate on. Example: **ipconfig /all** where **/all** is the argument.*

What do defenders inspect command-line arguments?

They often reveal what a program was actually doing, whether activity was normal or malicious, and the attacker's intent and techniques.

Windows Defender + Security Features

Open Windows Security.

What is Microsoft Defender?

Microsoft's built-in security platform designed to protect Windows systems from: Malware, Viruses, Ransomware, Phishing, Hackers, and Suspicious activity. It scans files, monitors processes, checks downloads, watches network activity, inspects scripts, and detects suspicious behavior.

What is real-time protection?

A feature in Microsoft Defender and other antivirus systems that continuously monitors a computer for threats while the system is running.

What is attack surface reduction?

A security approach focused on reducing the number of ways an attacker can compromise a system.

How do defenders correlate logs with network activity?

Comparing events from multiple systems and matching them using shared details like timestamps, IP addresses, usernames, hostnames, processes, and ports.